

CSE421: Computer Networks

Chapter 4: The Transport Layer — TCP Part II

(Reliability, Error Control, & Flow Control)

Exam Study Guide

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1 Reliability in TCP: Core Mechanisms

TCP guarantees reliable delivery over an inherently unreliable IP network. When a receiver receives segments, it performs verification, error checks, and reassembly.

1.1 The Key Tasks of TCP Reliability

1. **Mathematical Integrity Check:** Every segment is mathematically checked at the destination to ensure it was not corrupted during transit.
2. **Handling Out-of-Order Segments:** If segments arrive out of order, TCP stores them in a temporary receiver buffer and reassembles them in their correct order based on sequence numbers.
3. **Eliminating Duplicates:** If a network path duplicates a packet, the receiver identifies it via duplicate sequence numbers and discards the extra segment.
4. **Preserving Sequence Numbers on Empty ACKs:** When a TCP segment contains no payload (it is just a pure acknowledgment message), **it does not consume a sequence number**. The sequence number in the header remains the same for the next actual data segment.

1.2 The Checksum Field

To detect bit-level corruption, TCP prepends a **Pseudo-Header** to the actual segment and calculates a 16-bit checksum.

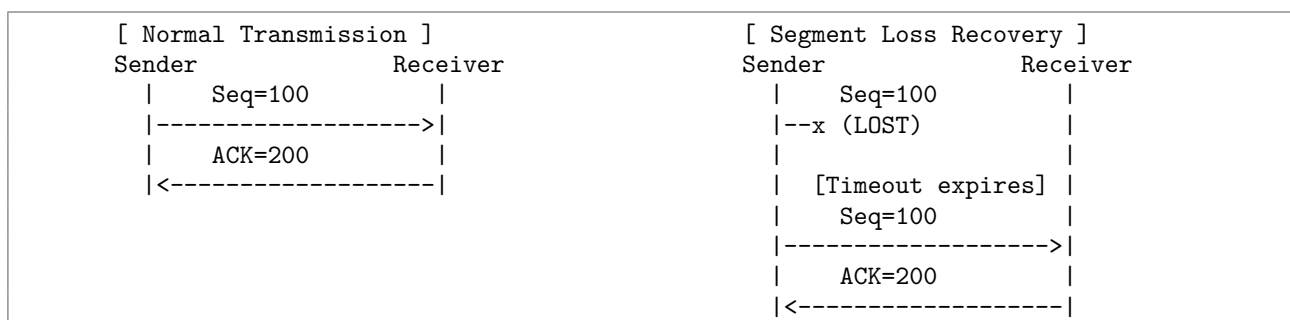
- **How it Works (1's Complement Sum):**

1. The sender blocks the data (including IP source/destination, protocol number, and TCP segment) into 16-bit words.
2. It sums these words using 1's complement arithmetic, and then takes the 1's complement of the total sum. This result is placed in the `Checksum` field.
3. The receiver performs the same sum over the received block (including the checksum). If the result is all 1s (or 0 depending on implementation), the data is intact. If any bit has flipped, the sum will fail, the receiver **silently drops** the corrupted segment, and waits for a retransmission.

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2 Error Control

TCP uses sequence numbers, cumulative acknowledgments, and timers to track and recover lost or corrupted packets.



2.1 Timeout and Retransmission (RTO)

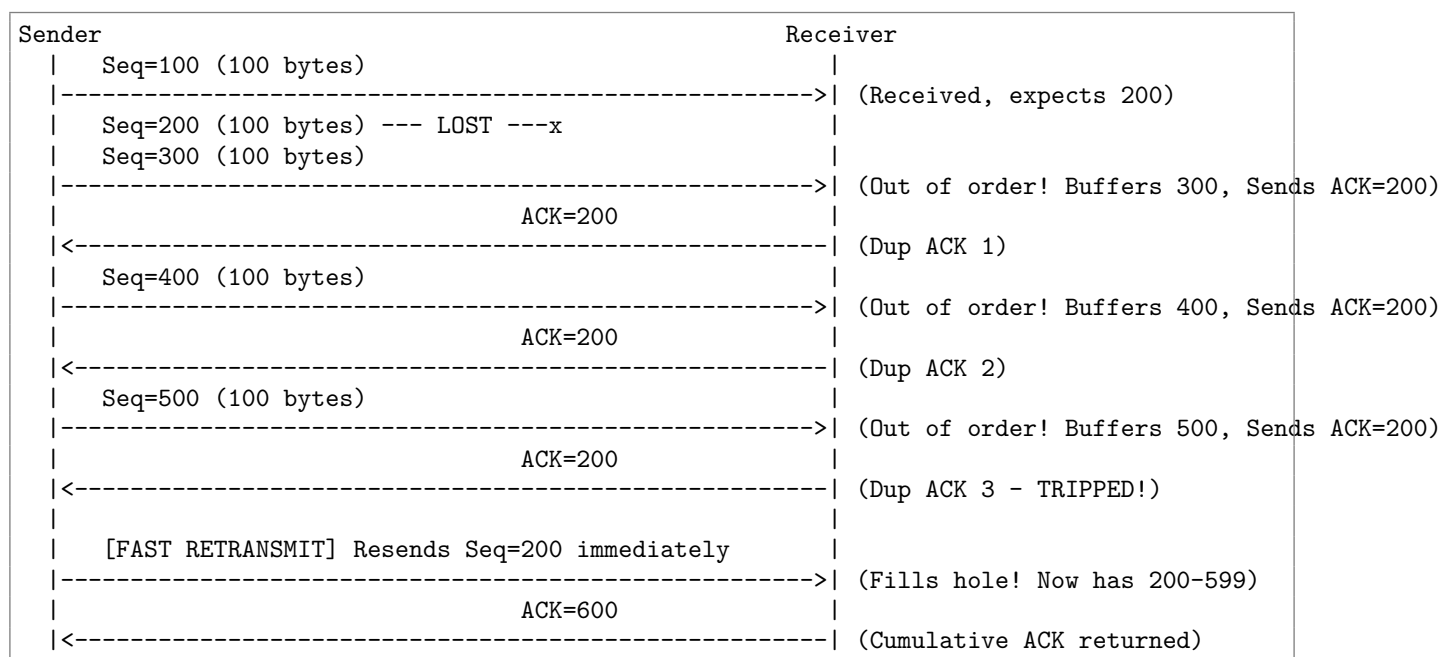
- **Retransmission Timeout (RTO):** Every time TCP sends a segment, it starts a retransmission timer. The duration of this timer is dynamically calculated based on the Round-Trip Time (RTT).
- **Trigger:** If the timer expires before the sender receives an ACK for that segment, TCP assumes the segment was lost in transit and **retransmits** it.

2.2 Fast Retransmit: The 3-Duplicate ACKs Rule

Waiting for a timer to expire (RTO) takes a relatively long time and hurts throughput. TCP uses an optimization called **Fast Retransmit**.

- **The Mechanism:**

1. If a segment is lost, any subsequent out-of-order segments that successfully reach the receiver will trigger immediate ACKs.
2. These ACKs will all request the sequence number of the *missing* segment (duplicate ACKs).
3. When the sender receives **three duplicate ACKs** for the same sequence number (meaning four identical ACKs in total), it is highly likely the segment was lost and not just delayed.
4. The sender immediately retransmits the missing segment **before its timer expires**.



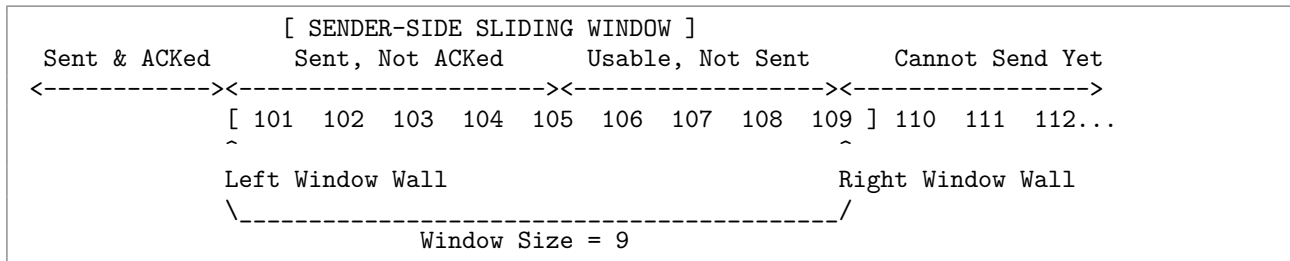
3 Flow Control: Sliding Window

Flow Control prevents a fast sender from overwhelming a slow receiver. TCP uses a **Sliding Window** protocol.

3.1 Buffer State vs. Window Size

- The receiver allocates a buffer to store incoming data before the application reads it.
- The receiver advertises its free space using the `Window Size` field in every ACK.

- **Formula:**

$$\text{Advertised Window (rwnd)} = \text{Max Receive Buffer} - (\text{Last Byte Received} - \text{Last Byte Read})$$


3.2 Detailed Trace Example of the Sliding Window

Imagine a system with **Sender (S)** and **Receiver (R)**:

- Initial conditions: **Window Size = 300 bytes.**
- Initial Sequence Number (**Seq = 1000**).

1. **S sends Segment 1:** Seq = 1000, size = 100 bytes (Bytes 1000-1099).
2. **S sends Segment 2:** Seq = 1100, size = 100 bytes (Bytes 1100-1199).
3. **S sends Segment 3:** Seq = 1200, size = 100 bytes (Bytes 1200-1299).

- *Sender's buffer state:* 300 bytes are outstanding (sent, not acknowledged). The sliding window is completely full; S cannot transmit any more bytes.

4. R receives Segment 1 and Segment 2, but Segment 3 is delayed.

5. **R sends an ACK:** ACK = 1200 (meaning it got up to 1199), and advertises a Window Size = 200 (R's application hasn't read the data yet, so the buffer space shrank).

- 6. S receives ACK = 1200, Win = 200:**

- S deletes bytes 1000-1199 from its local retry buffer.
- The left wall of S's window slides up to 1200.
- With $Win = 200$, S's window now covers bytes 1200 to 1399.
- Since segment 3 (Bytes 1200-1299) is already sent but unacknowledged, S has exactly $200 - 100 = 100$ bytes of usable window remaining. S can now transmit another 100-byte segment (e.g., Seq = 1300).

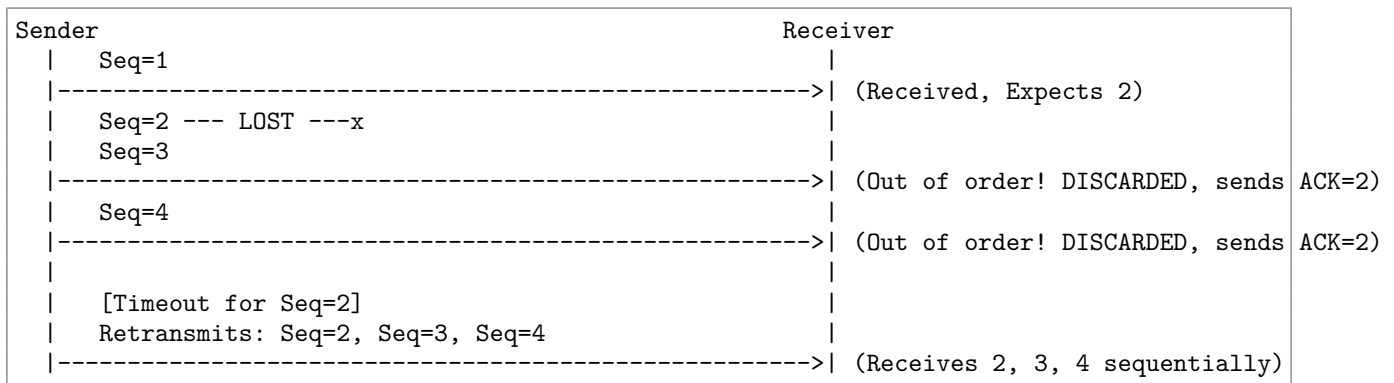
4 Retransmission Protocols: Go-Back-N vs. Selective Repeat

When packets are lost, the sender's behavior changes dramatically depending on whether **Go-Back-N (GBN)** or **Selective Repeat (SR)** is used.

4.1 1. Go-Back-N (GBN)

GBN is a simple protocol where the receiver has an extremely limited buffer size (often equal to 1).

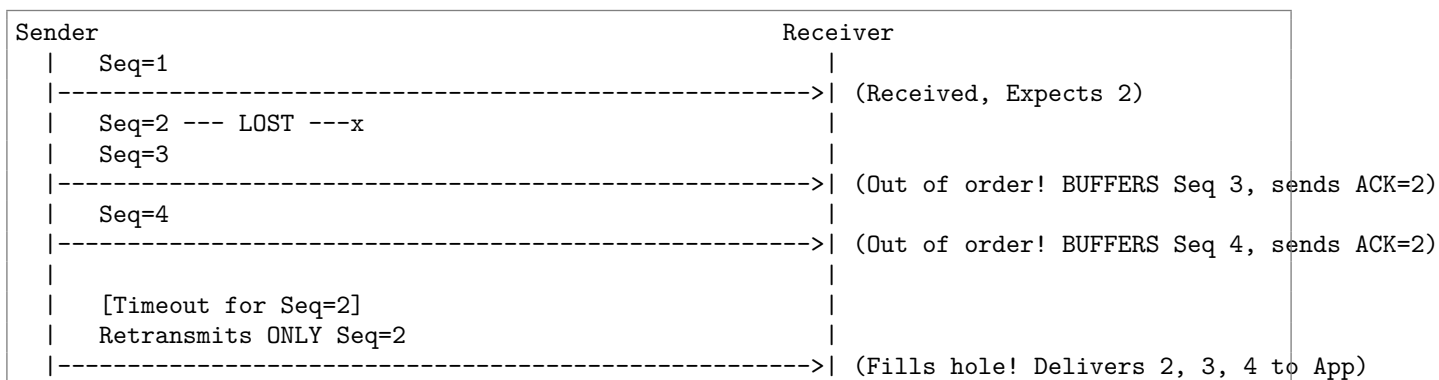
- **Receiver Rules:** The receiver **only accepts segments in order**. If any segment is missing, the receiver discards all subsequent out-of-order segments without saving them.
- **Sender Rules:** The sender maintains a single timer for the oldest unacknowledged packet. If a timeout occurs, the sender **retransmits all outstanding packets** starting from the lost one (it “goes back N” steps).



4.2 2. Selective Repeat (SR) / Selective ACK (SACK)

Selective Repeat is a more advanced protocol that maximizes bandwidth utilization by avoiding unnecessary retransmissions.

- **Receiver Rules:** The receiver accepts and **buffers out-of-order segments** as long as they fall within its receiving window. It sends specific selective ACKs for those buffered segments.
- **Sender Rules:** The sender maintains individual timers for each transmitted segment. When a timeout occurs (or 3-Duplicate ACKs are received), **the sender only retransmits the specific segment that was lost**.



5 Summary Comparison for the Exam

Metric	Go-Back-N (GBN)	Selective Repeat (SR) / SACK
Receiver Buffer Size	1 (Discards out-of-order)	> 1 (Stores out-of-order)
Retransmitted Segments	All outstanding packets starting from the lost one.	Only the single lost packet.
Complexity	Low complexity, low memory overhead.	High complexity, requires buffering at both sender and receiver.
Bandwidth Efficiency	Poor on high-loss links.	Highly efficient.

Table 1: Comparison of Go-Back-N and Selective Repeat